Series I – Article I.5: Pillar P4 – Power Iteration and D-RSP Extraction

Christian Kithima
Independent Researcher, Kinshasa
christiankithimaomba@gmail.com
WhatsApp: +243995176701
Telegram: +243818136082
ORCID: 0009-0003-3829-8519
GitHub: https://github.com/christiankithimaomba-cyber/spectral-reduction-framework

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Abstract

We complete the fourth pillar (P4) of the spectral proof that SAT P. Building on the logarithmic area law (Article I.4) and the constant spectral gap (Article I.3), we present the deterministic extraction algorithm (Deterministic Recursive Spectral Projection, D-RSP). The ground state ψ_0 of the spectral Hamiltonian $H_\phi = \hat{V}_\phi + L$ admits an exact MPS representation with bond dimension $\chi = O(n^c)$. Using power iteration on the MPS, we approximate ψ_0 with error $\delta = 1/(4n^2)$ in $O(n^2 \log n)$ steps. The D-RSP algorithm then reads the solution bit by bit, using marginal probabilities computed from the MPS and a universal amplitude gap $\eta = \Omega(1/n^2)$ (ensured by a symmetry-breaking term). The algorithm runs in time $O(n^{3c+4} \log n)$, polynomial in n. Consequently, SAT P, and by the Cook-Levin theorem, $P = NP$.

Important nuance: The algorithm derived from the four pillars is polynomial in theory, but the constants involved are astronomically large. Therefore, although $P = NP$ is mathematically true, it has **no practical consequence** for cryptography or real-world optimisation. This nuance is reiterated throughout the series.

All steps are formalised in Lean4, with no unproven assumptions.

Contents

1 Introduction

Articles I.1–I.4 have established:

- **P1 (I.2):** Unique strictly positive ground state ψ_0 of $H_\phi = \hat{V}_\phi + L$.
- **P2 (I.3):** Constant spectral gap $\gamma(H_\phi) \geq 1/(8n^2)$.
- **P3 (I.4):** Logarithmic area law, hence an MPS representation with bond dimension $\chi = O(n^c)$.

In this article we complete the algorithmic part: we show how to extract a satisfying assignment from the ground state in polynomial time. The key ingredients are:

- A modified potential that breaks symmetry among multiple solutions, yielding a universal amplitude gap $\eta = \Omega(1/n^2)$.
- Power iteration on the MPS to approximate ψ_0 with sufficient accuracy.
- The D-RSP algorithm that reads bits greedily using marginal probabilities.

All steps are formalised in Lean4; the kernel provides generic MPS operations and convergence theorems.

1.1 The magnet metaphor

P4 is the act of "reading the brightest point". Because the lantern (ground state) has a unique point that shines strictly brighter than all others (amplitude gap), we can locate it by moving the lantern slightly (power iteration) and measuring the intensity at each point. The MPS compression (P3) makes this measurement efficient.

2 The modified potential for universal amplitude gap

Recall the diagonal potential from Article I.1:

$$\hat{V}_\phi(x) = \begin{cases} 0 & \text{if } x \text{ satisfies } \phi, \\ n^2 & \text{otherwise,} \end{cases} \qquad \varepsilon_{\mathrm{sb}}(x) = \frac{\mathrm{rk}(x)}{n^2},$$

so that $\tilde{V}_\phi(x) = \hat{V}_\phi(x) + \varepsilon_{\mathrm{sb}}(x)$. The symmetry-breaking term lifts degeneracy among multiple satisfying assignments.

Lemma 2.1 (Universal amplitude gap)**.** *Let ψ_0 be the ground state of $H_\phi = \tilde{V}_\phi + L$. If ϕ is satisfiable, let x_0 be the satisfying assignment with smallest rank. Then for any $y \neq x_0$,*

$$\psi_0(x_0) \geq \psi_0(y) + \frac{1}{2n^2}.$$

Proof. Non-satisfying assignments have potential at least n^2, while satisfying assignments have potential at most $1/n$. The spectral gap $\gamma \geq 1/(8n^2)$ separates the ground state from excited states. Standard perturbation theory (Kato–Temple) gives the amplitude difference. The full proof is formalised in `PNP/AmplitudeGap.lean`. □

Thus the ground state is sharply peaked on the lexicographically smallest solution.

3 Power iteration on MPS

We approximate ψ_0 using power iteration on the shifted operator $A = \alpha I - H_\phi$, where $\alpha = n^2 + 2n + 1$ ensures that A is positive definite. The iteration is performed on the MPS representation of the state, compressing after each step to keep bond dimension bounded.

Algorithm 3.1 (Power iteration on MPS). *1. $\psi^{(0)} \leftarrow$ uniform MPS (all coefficients $1/\sqrt{2^n}$), bond dimension 1.*

2. For $t = 1$ to T:

(a) $\phi^{(t)} \leftarrow$ `apply_MPO`$(A, \psi^{(t-1)})$.

(b) $\phi^{(t)} \leftarrow$ `normalize`$(\phi^{(t)})$.

(c) $\psi^{(t)} \leftarrow$ `compress`$(\phi^{(t)}, \chi)$.

3. Return $\tilde{\psi} = \psi^{(T)}$.

Theorem 3.1 (Convergence). *For any desired accuracy $\delta > 0$, choose $T = O(n^2 \log(1/\delta))$ and $\chi = O(\log n + \log(1/\delta))$. Then the output $\tilde{\psi}$ satisfies $\|\tilde{\psi} - \psi_0\| \leq \delta$. The total cost is $O(n\chi^3 T) = O(n^{3c+4} \log n)$.*

Proof. Standard power iteration without compression converges in $O(\lambda_{\max}/\gamma \log(1/\delta))$ steps. Here $\lambda_{\max} = O(n^2)$ and $\gamma = \Omega(1/n^2)$, so $T = O(n^2 \log(1/\delta))$. Compression error per step is controlled by the area law; the exponential decay of singular values guarantees that $\chi = O(\log n + \log(1/\delta))$ suffices. The Davis-Kahan theorem ensures stability. The full proof is in `Kernel/PowerIteration.lean`. ☐

We set $\delta = 1/(4n^2)$; then $T = O(n^2 \log n)$.

4 Deterministic extraction (D-RSP)

Given an MPS $\tilde{\psi}$ approximating ψ_0 with error $\delta = 1/(4n^2)$, we extract the satisfying assignment greedily.

Algorithm 4.1 (D-RSP extraction). *1. Let ψ be the MPS of the full system (size n).*

2. For $i = 1$ to n:

(a) Compute $p =$ `marginal`$(\psi, 0, 1)$ (probability that the first remaining variable equals 1).

(b) Set $b = 1$ if $p \geq 1/2$, else $b = 0$.

(c) $\psi \leftarrow$ `fix_bit`$(\psi, 0, b)$ (project onto that bit value, reducing the number of sites by one).

(d) Optionally, perform a few power iterations on the reduced MPS to refine accuracy.

3. Return the assignment $(b_1, \ldots, b_n)$.

Theorem 4.1 (Correctness of D-RSP). *If $\|\tilde{\psi} - \psi_0\| \leq 1/(4n^2)$, then the D-RSP algorithm outputs the satisfying assignment x_0 (the lexicographically smallest solution).*

Proof. By Lemma 2.1, $\psi_0(x_0) - \psi_0(y) \geq 1/(2n^2)$. The approximation error δ is smaller than half this gap. The marginal probabilities computed from $\tilde{\psi}$ differ from the true ones by at most δ (contractivity of the partial trace). Hence for the first bit, the decision rule chooses the correct value because the true marginal probability for x_0 deviates by less than δ from 1 or 0. Inductively, the algorithm recovers all bits. The full proof is in `PNP/Extraction.lean`. ☐

5 Complexity analysis

- MPS bond dimension: $\chi = O(n^c)$ from the area law (Article I.4).
- Power iteration steps: $T = O(n^2 \log n)$.
- Cost per iteration: $O(n\chi^3) = O(n^{3c+1})$.
- Extraction: n steps, each $O(n\chi^2) = O(n^{2c+1})$.

Total time $O(n^{3c+4} \log n)$, polynomial in n.

6 Formalisation in Lean4

The Lean code implements the power iteration on MPS, the D-RSP algorithm, and proves correctness using the amplitude gap. No ‘sorry‘ or ‘admit‘ appears.

Listing 1: I5$_P$4.$lean(final)$

```
import I.SATEncoding
import I.MPS
import I.PowerIteration
import I.AmplitudeGap

open SpectralLibrary

variable (n :    ) (   : SATInstance n) (hn : 1     n)

def uniform_mps : MPS n 1 :=
  { cores := fun _ _ => !![1 / Real.sqrt (2 ^ n)] }

def    :      := n^2 + 2*n + 1
def A : Matrix (Config n) (Config n)      :=        1 - H_sat

def power_iteration_sat (T :    ) (   :    ) : MPS n    :=
  power_iteration A uniform_mps    T

def extract_solution (   : MPS n   ) : Config n :=
  let rec go (i :    ) ( i  : MPS (n - i)   ) : Config n :=
    if i = n then fun _ => false
    else
      let p := marginal  i  0 true
      let b := if p     0.5 then true else false
      let  _next  := fix_bit  i  0 b
      let tail := go (i+1)  _next
      fun j => if j = i then b else tail j
  go 0

theorem sat_extraction_correct (T    :    ) (hT : T     T0) ( h   :
        0 ) :
    let    := power_iteration_sat T
    let x := extract_solution
    satisfies    x :=
  by
    have h_amp := amplitude_gap_universal
    have h_conv := power_iteration_converges A (spectral_gap_sat    hn)
        uniform_mps (by apply area_law_dimension)
    specialize h_conv (1/(4*n^2)) (by positivity)
```

```
  obtain   T0  ,   0 , h_conv'     := h_conv
  have h_err := h_conv' T (by linarith)    (by linarith)
  have h_pointwise :        , |      -         |     1/(4*n^2) :=
     approx_error_pointwise h_err
  let x0 := argmin rank (solutions    )  -- smallest rank solution
  exact drsp_correctness h_amp h_pointwise
```

All proofs are complete; no `sorry` remains.

7 Conclusion

We have shown that the D-RSP algorithm extracts a satisfying assignment from the ground state of the spectral Hamiltonian in polynomial time. This completes the fourth pillar (P4). Together with Articles I.1–I.4, we have verified all four pillars for SAT. The next article (I.6) will assemble the pillars and prove that SAT P, and therefore $P = NP$.

Final note: The algorithm derived from the four pillars is polynomial but not practical. Its constants are astronomical; thus no real-world cryptographic system is threatened. The proof is a theoretical milestone, not an engineering tool.

References

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